Datamites

Global Institute for Data Science

HUMAN RESOURCE (HR) ANALYSIS

POWER BI DASHBOARD
CAPSTONE PROJECT REPORT

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ABSTRACT

Human Resource Analytics is the process of collecting, analyzing, and interpreting workforce data to improve organizational decision-making and employee management. Organizations generate large amounts of employee-related information every day, including recruitment records, employee demographics, work locations, department structures, and workforce performance metrics.

The objective of this project is to analyze employee data and identify workforce trends using data analytics and visualization techniques. This project focuses on understanding employee demographics, hiring patterns, department distribution, work location preferences, and workforce diversity.

Using Power BI, interactive dashboards were developed to transform raw employee data into meaningful insights. The dashboard enables HR departments to monitor workforce metrics efficiently and supports strategic planning for employee management.

This project demonstrates how business intelligence tools can improve workforce planning, increase operational efficiency, and support data-driven decision-making

INTRODUCTION

Human Resource Management plays a critical role in organizational success because employees are one of the most valuable assets of any organization. HR departments manage various functions including recruitment, employee records, workforce planning, employee engagement, and resource allocation.

As organizations grow, employee data volume increases significantly, making manual analysis difficult and time-consuming.

HR Analytics helps organizations:

* Analyze employee demographics

* Understand workforce distribution

* Monitor hiring patterns

* Track remote and office employees

* Improve workforce planning

* Support strategic decisions

The purpose of this project is to analyze workforce-related data and provide insights through visual dashboards. The project transforms employee information into actionable business intelligence..

PROBLEM STATEMENT

Organizations often face several challenges while managing employee information.

Common challenges include:

* Large employee datasets that are difficult to analyze manually
* Difficulty tracking hiring trends over time
* Challenges in understanding workforce distribution
* Lack of visibility into remote and headquarters workforce
* Limited insights into diversity and department allocation

Without analytics solutions:

* Decision-making becomes slower
* Workforce planning becomes inefficient
* Employee distribution issues remain unidentified
* HR operations become less effective

Therefore, organizations require analytics-driven solutions that help simplify workforce monitoring and improve decision quality.

PROJECT OBJECTIVES

The objectives of this HR Analytics project are divided into operational and business goals.

Primary Objectives:

* Analyze workforce demographics
* Calculate total employee count
* Identify percentage of remote employees
* Identify percentage of HQ employees
* Analyze employee distribution across departments
* Study workforce diversity

Secondary Objectives:

* Develop interactive dashboards
* Generate workforce insights
* Support HR decision-making

Business Objectives:

* Improve workforce planning
* Enhance resource allocation
* Monitor recruitment effectiveness
* Support organizational growth strategies

DATASET DESCRIPTION

The dataset contains employee-related information collected from organizational records.

Important columns in the dataset include:

Employee ID:

Unique identifier assigned to every employee.

First Name / Last Name:

Employee personal information.

Gender:

Employee gender classification.

Race:

Employee ethnicity classification for diversity analysis.

Age:

Employee age information.

Department:

Department where employees are assigned.

Job Title:

Employee role designation.

State:

Employee working location.

Hire Date:

Date employee joined the organization.

Birth Date:

Used for age calculation.

Location Type:

Determines whether employees work remotely or from headquarters.

Employment Status:

Active or inactive employee status.

The dataset provides sufficient information for analyzing workforce patterns and employee distribution.

TOOLS AND TECHNOLOGIES USED

Power BI:

Power BI was used as the primary analytics tool because of its interactive dashboard capabilities.

Applications:

* Dashboard development
* KPI creation
* Data visualization
* Report generation

Power Query:

Used for:

* Data cleaning
* Transformation
* Handling missing values

DAX:

Used for:

* Measures
* Calculated columns
* Percentage calculations
* KPI metrics

METHODOLOGY / PROJECT PROCESS

Phase 1: Data Collection

Collected employee records from available datasets.

Activities:

* Imported dataset into Power BI

* Verified column structure

* Checked dataset quality

Phase 2: Data Understanding

Performed:

* Column inspection

* Data type validation

* Missing value analysis

Phase 3: Data Cleaning

Cleaning activities:

* Removed duplicates

* Corrected incorrect values

* Fixed date columns

* Handled null values

Phase 4: Data Transformation

Created:

* Remote worker column

* Headquarters worker column

* Age calculations

* KPI measures

Phase 5: Dashboard Development

Developed:

* KPI Cards

* Charts

* Filters

* Visual dashboards

Phase 6: Insight Generation

Performed workforce analysis and generated recommendations.

DATA CLEANING AND PREPROCESSING

Data preprocessing ensures data quality and reliability.

Steps performed:

Step 1:

Removed duplicate employee records.

Step 2:

Handled missing values.

Step 3:

Corrected date formatting issues.

Step 4:

Changed incorrect data types.

Step 5:

Created new columns for:

* Remote workers

* HQ workers

* Employee age

Step 6:

Validated transformed data.

Benefits:

* Improved analysis accuracy

* Better dashboard performance

* Reduced reporting errors

DASHBOARD DEVELOPMENT

The dashboard was designed to provide quick workforce insights.

KPIs Created:

* Total Employees

* Average Employee Age

* Percentage Remote Workers

* Percentage HQ Workers

Charts Created:

Line Chart:

Hiring Rate by Year

Purpose:

Analyze workforce growth trends.

Donut Chart:

Gender Distribution

Purpose:

Study workforce diversity.

Bar Charts:

Employees by State

Purpose:

Employees by Department

Purpose:

Identify department workforce allocation.

Employees by Race

Purpose:

Analyze diversity metrics.

Slicers Added:

* Department

* State

* Gender

* Work Location

Dashboard Benefits:

* Easy workforce monitoring

* Better reporting

* Interactive exploration

KEY INSIGHTS

Insights obtained from analysis:

* Workforce concentration is higher in certain departments.

* Remote work adoption has increased significantly.

* Some states contain larger employee populations.

* Hiring rates fluctuate over different years.

* Workforce diversity varies among departments.

* Gender distribution is uneven across teams.

These findings help HR departments improve workforce planning.

RECOMMENDATIONS

Based on analysis, the following recommendations were developed:

1. Improve workforce balance across departments.
2. Increase employee diversity initiatives.
3. Monitor remote employee engagement.
4. Improve workforce forecasting.
5. Strengthen recruitment planning.
6. Develop better retention strategies.

Expected benefits:

* Improved workforce utilization
* Better employee management
* Increased efficiency
* Stronger HR decision-making

CONCLUSION

The HR Analytics project successfully converted employee data into actionable business insights.

Through visualization and data analysis, workforce patterns, hiring trends, and employee distribution were identified.

Power BI dashboards enabled effective workforce monitoring and supported strategic decision-making.

The project demonstrates the importance of analytics in improving HR operations and organizational planning.

FUTURE SCOPE

Potential future improvements include:

* Predictive attrition analysis

* Machine learning integration

* Workforce forecasting

* Real-time employee monitoring dashboards

* Employee performance analytics

* Automated HR reporting systems

REFERENCES

1. Microsoft Power BI Documentation
2. Power Query Documentation
3. DAX Documentation
4. Datamites Internship Materials
5. Employee Dataset Used for Analysis